UNIVERSITÄT
LEIPZIG

Natural Language Processing

Week 1:Introduction and Organisation

Leonie Weissweiler 09.04.2026

About Me

- Assistant Professor (Juniorprofessorin) for Natural Language Processing
- Can be emailed in either German or English
- No fixed office hours, please email
- New here, still figuring out administrative issues

Basic Information: Lecture

Lecture

- Where: Felix-Klein-Hörsaal
- When: Thursdays 9-11 c.t.
- Who: Leonie Weißweiler
- Moodle: only Vorlesung will be used, ignore the Übung pages
- Slides will be uploaded to moodle after each lecture

Basic Information: Tutorials

Tutorials

- Where: SG 3-12
- When: Wednesdays, either 11-13 c.t. or 15-17 c.t.
- Who: Toshiki Nakai, Jacob Suchardt
- Exercise sheet will be uploaded to moodle after each lecture
- Exercise sheet is voluntary and will be discussed after 6 days in the tutorials
- Questions will be exam-style but may eventually also include programming

Course Language

- English
- If you must, you can email me in German but email the TAs in English

Course Prerequisites

- Basic familiarity with python is useful for following along with programming exercises

(Preliminary) Semester Plan

- 09.04. Intro and NLP problems
- 16.04. Text processing
- 23.04. Corpora and tokenisation
- 30.04. N-gram language models
- 07.05. Transformers 1
- 21.05. Transformers 2
- 28.05. LLMs 1
- 04.06. LLMs 2
- 11.06. Morphology
- 18.06. Syntax
- 25.06. Semantics
- 02.07. Pragmatics
- 09.07. Outlook: Current Issues

Exam

- Week after the semester ends, exact date tbd
- Written, 90 minutes
- Similar to exercise sheets
- Second exam (Nachklausur) near the beginning of the winter semester tbd

Questions about organisation?

Today's Lecture: Introduction and NLP Problems

Language and Technology

- **Computational Linguistics (CL)** is the study of using computational models to understand and process human language.
- **Natural Language Processing (NLP)** is a field at the intersection of linguistics and artificial intelligence that focuses on enabling computers to understand, generate, and manipulate human language.

Computational Linguistics vs Natural Language Processing

- Science
- Which theory is correct?

- Engineering
- Which theory is useful?

Related Fields

- Linguistics
- Statistics
- Psychology
- Machine Learning and AI

Textbooks

- Eisenstein: Natural Language Processing. MIT Press 2019 https://github.com/jacobeisenstein/gt-nlp-class/tree/master/notes
- Jurafsky & Martin: Speech and Language Processing. Prentice Hall 2000, 2009, 2025. https://web.stanford.edu/~jurafsky/slp3/
- Goldberg: Neural Network Methods for Natural Language Processing. Morgan & Claypool Publishers 2017.

More Applied Textbooks

- Rao & McMahan: Natural Language Processing with PyTorch. O'Reilly 2019. https://github.com/delip/PyTorchNLPBook
- Tunstall, Werra & Wolf: Natural Language Processing with Transformers. O'Reilly 2022. https://github.com/nlp-with-transformers

Scientific Venues

- Association for Computational Linguistics (ACL)
- Empirical Methods in Natural Language Processing (EMNLP)
- Regional ACLs: EACL, NAACL, AACL
- Non-ACL conferences. COLING, LREC
- Journals: Transactions of the ACL (TACL), Computational Linguistics (CL)
- All papers can be found in the ACL anthology: https://aclanthology.org/

Software Packages

- Main (maybe only) programming language is Python
- Important packages to keep in mind for later:
  - spaCy
  - Stanza
  - NLTK
  - Scikit-learn
  - Hugging Face

Goals of Natural Language Processing

- 1. Aid humans in writing.

  Correcting mistakes, formulating and paraphrasing text, transcription.

- 2. Identify texts related to spoken or written requests.

  Text information retrieval, semantic text similarity, question answering.

- 3. Make sense of texts without reading the originals.

  Categorization, information extraction, summarization, translation.

- 4. Instruct and be advised by a computer.

  Audio interfaces (e.g., dialog systems, robotics), learning and assessment.

- 5. Converse with computers as if they were human.

  Turing test, Conversational AI and chatbots, argumentation, affective computing.

- 6. Understanding the nature of language and its relation to (artificial) intelligence.

  Pragmatics, causality, alignment and grounding, multimodality, simulation, linguistics.

Examples of NLP Systems

Writing: Spelling and Grammar Checking

Alan Turing

“Alan Mathison Turing (23 June 1912 – 7 June 1954) was an english mathematician, computer scientist, logician, cryptanalyst, philosopher and theoretical biologist. Turing was highly influential in the developing of theoretical computer science, providing a formalisation of the concepts of algorithm and computatoin with the Turing machine, who can be considered a model of general-purpose computer. Turing is widely considered to been the farther of theoretical computer science and artificial intelligance. Despite these accomplishment he was ever fully recognised in his home country during his lifetime due to his homosexuality and because many of his work was covered by the Official Secrets Act.”

Can you spot any errors?

Writing: Spelling and Grammar Checking

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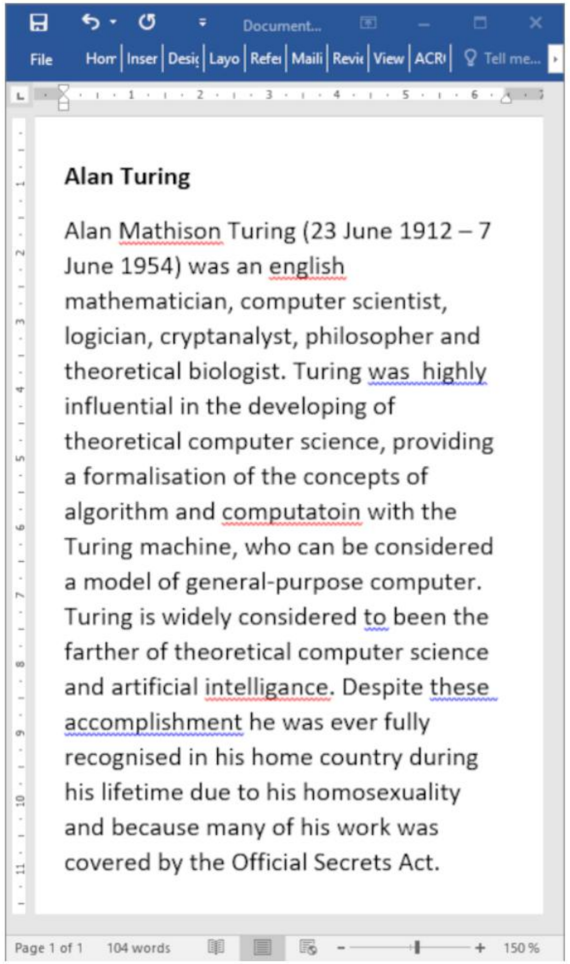
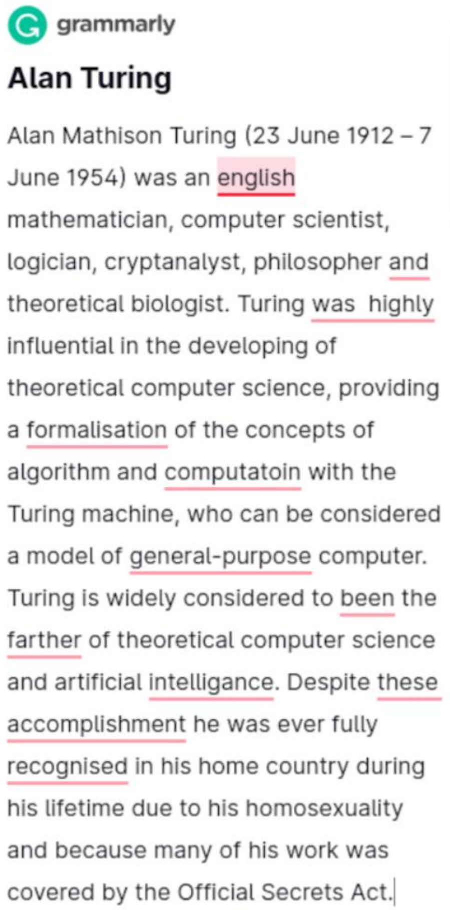
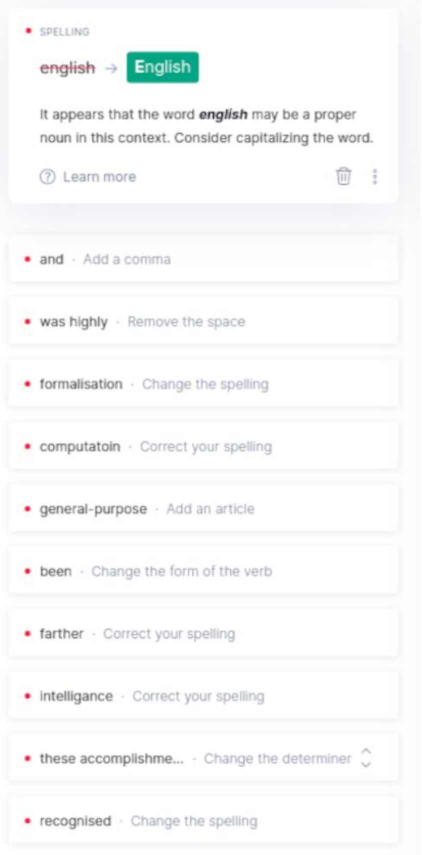
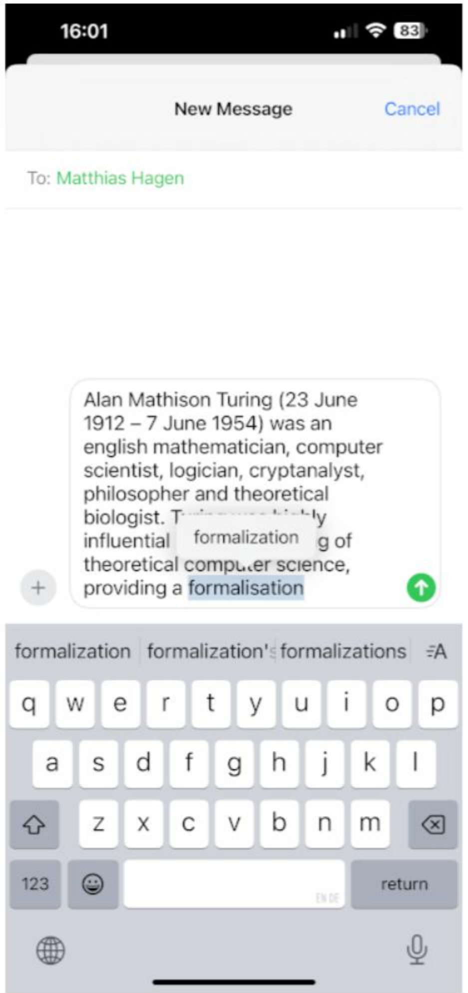
Can you spot any errors?

Writing: Spelling and Grammar Checking









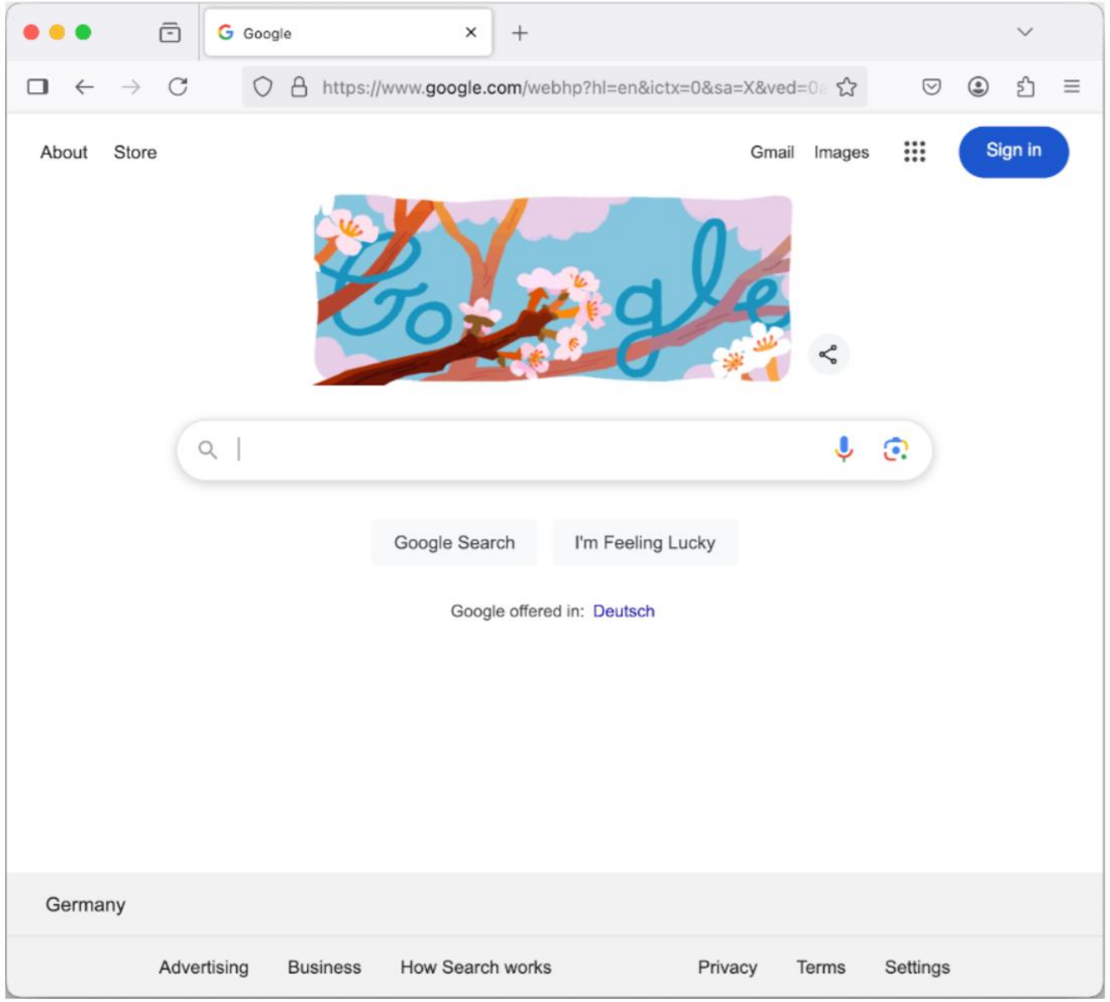
Remarks

- The text has been derived from a revision of the opening paragraph of the Alan Turing article on Wikipedia.
- Spelling and grammar checking includes detecting mistakes, highlighting them in the editor, and suggestingcorrections. It enables the writer to review their own mistakes, correct them manually, and learn from them.
- On mobile devices, writing is supported by autocompleting the current word, the next words, and optionallycorrecting typing mistakes automatically. This can lead to erroneous corrections.
- Detected errors:
  - english should be capitalized (Word, Grammarly)
  - and could be preceded by an Oxford comma (Grammarly)
  - was highly should only have one space (Word, Grammarly)
  - formalisation could be switched to American English spelling (all, dependent on locale)
  - computatoin should be ‘computation’ (all)
  - general-purpose should be preceeded by the article ‘a’ (Grammarly)
  - to been should be ‘be’ (Word, Grammarly; but Word for the wrong reason)
  - farther should be ‘father’ (Grammarly)
  - intelligance should be ‘intelligence’ (all)
  - these accomplishment should be ‘these accomplishments’ (Word, Grammarly)
  - recognised could be switched to American English spelling (all, dependent on locale)

False detections/auto-completions and undetected errors

- Mathison is correctly spelled; false positive (Word, meanwhile fixed)
- developing should be ‘development’; false negative (all)
- who should be ‘which’; false negative (all)
- farther should not be ‘farting’; false auto-completion (iPhone)
- ever should be ‘never’; false negative (all)
- many should be ‘much’; false negative (all)

Search


Google
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About
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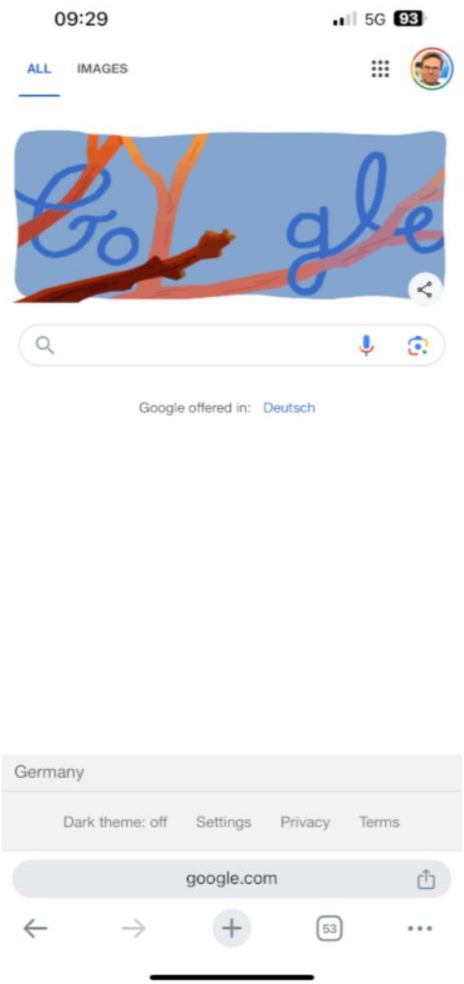
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Dark theme: off
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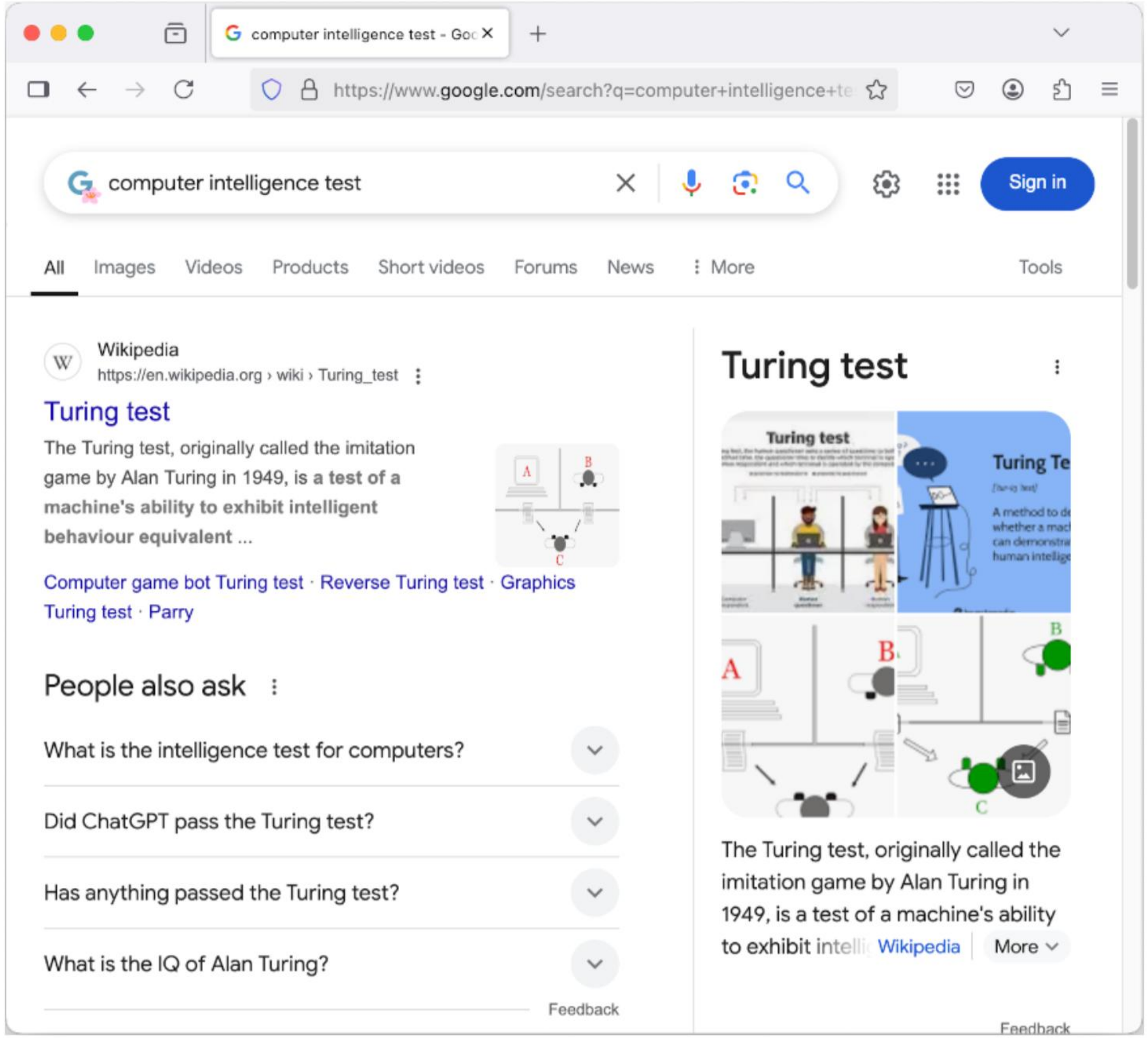
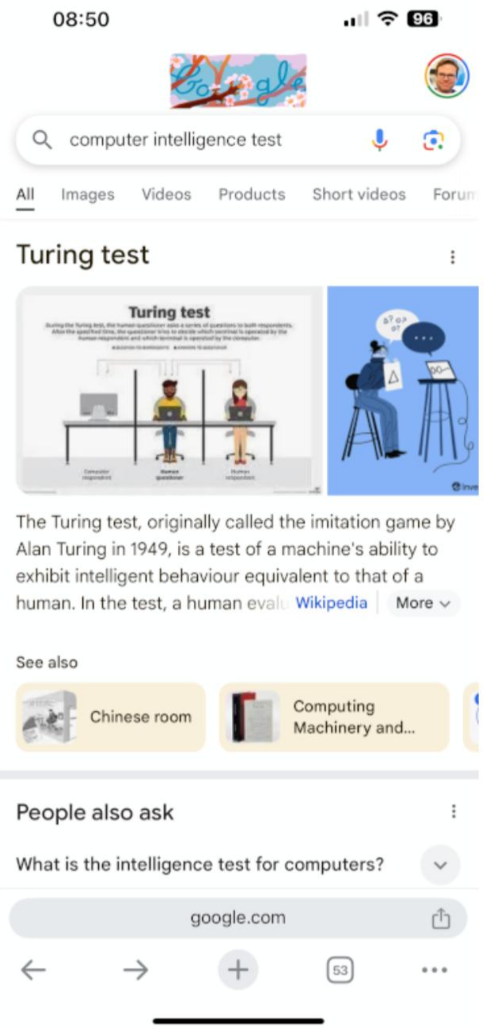
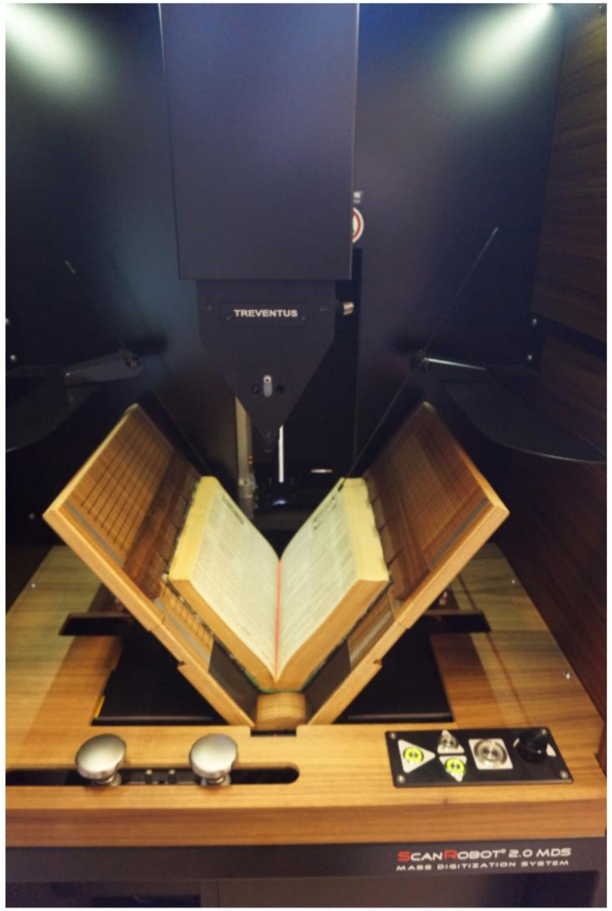
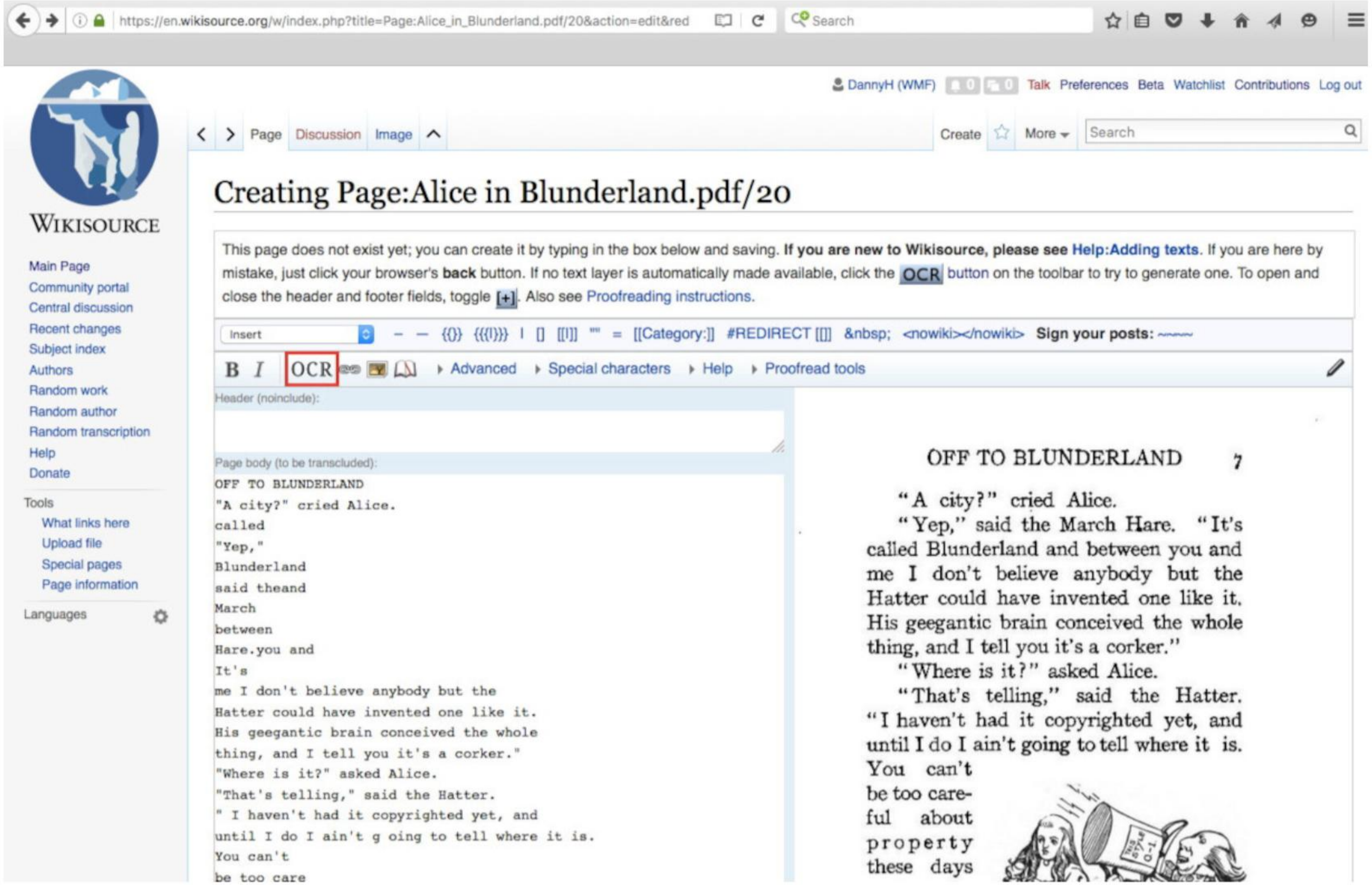
Search


computer intelligence test - Goo
https://www.google.com/search?q=computer+intelligence+te
computer intelligence test
Sign in
All
Images
Videos
Products
Short videos
Forums
News
More
Tools
Wikipedia
https://en.wikipedia.org › wiki › Turing_test
Turing test
The Turing test, originally called the imitation game by Alan Turing in 1949, is a test of a machine's ability to exhibit intelligent behaviour equivalent ...
Computer game bot Turing test · Reverse Turing test · Graphics Turing test · Parry
People also ask
What is the intelligence test for computers?
Did ChatGPT pass the Turing test?
Has anything passed the Turing test?
What is the IQ of Alan Turing?
Feedback
Turing test
The Turing test, originally called the imitation game by Alan Turing in 1949, is a test of a machine's ability to exhibit intelli
Wikipedia
More
Feedback

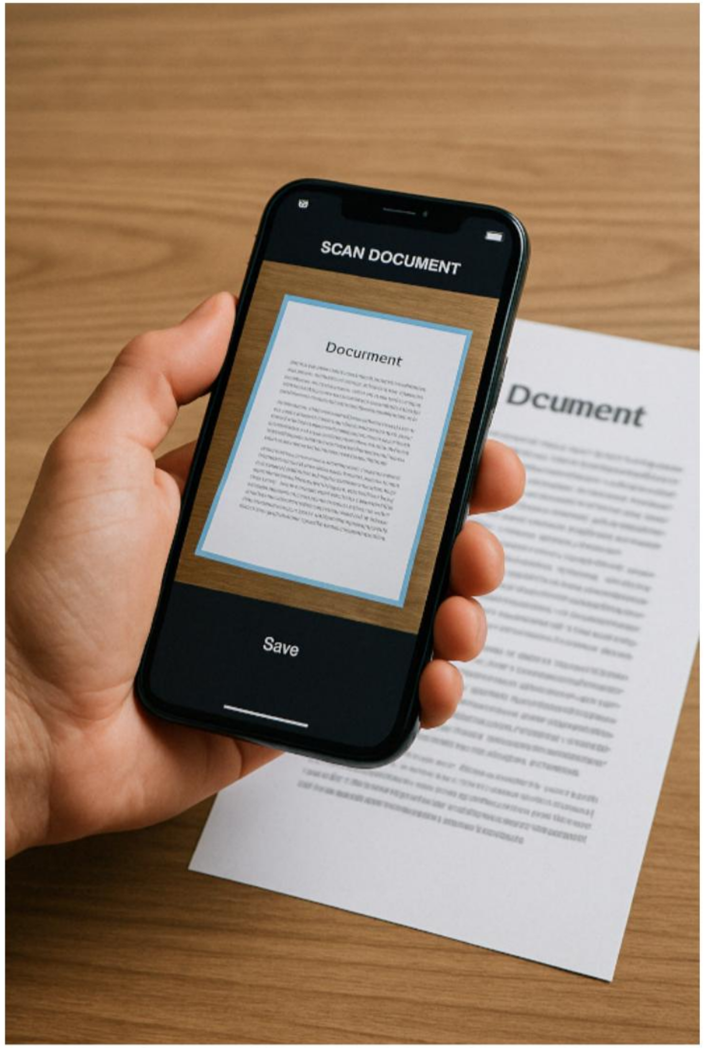

08:50
96
computer intelligence test
All
Images
Videos
Products
Short videos
Turing test
The Turing test, originally called the imitation game by Alan Turing in 1949, is a test of a machine's ability to exhibit intelligent behaviour equivalent to that of a human. In the test, a human evalu
Wikipedia
More
See also
Chinese room
Computing Machinery and...
People also ask
What is the intelligence test for computers?
google.com
53

Optical Character Recognition

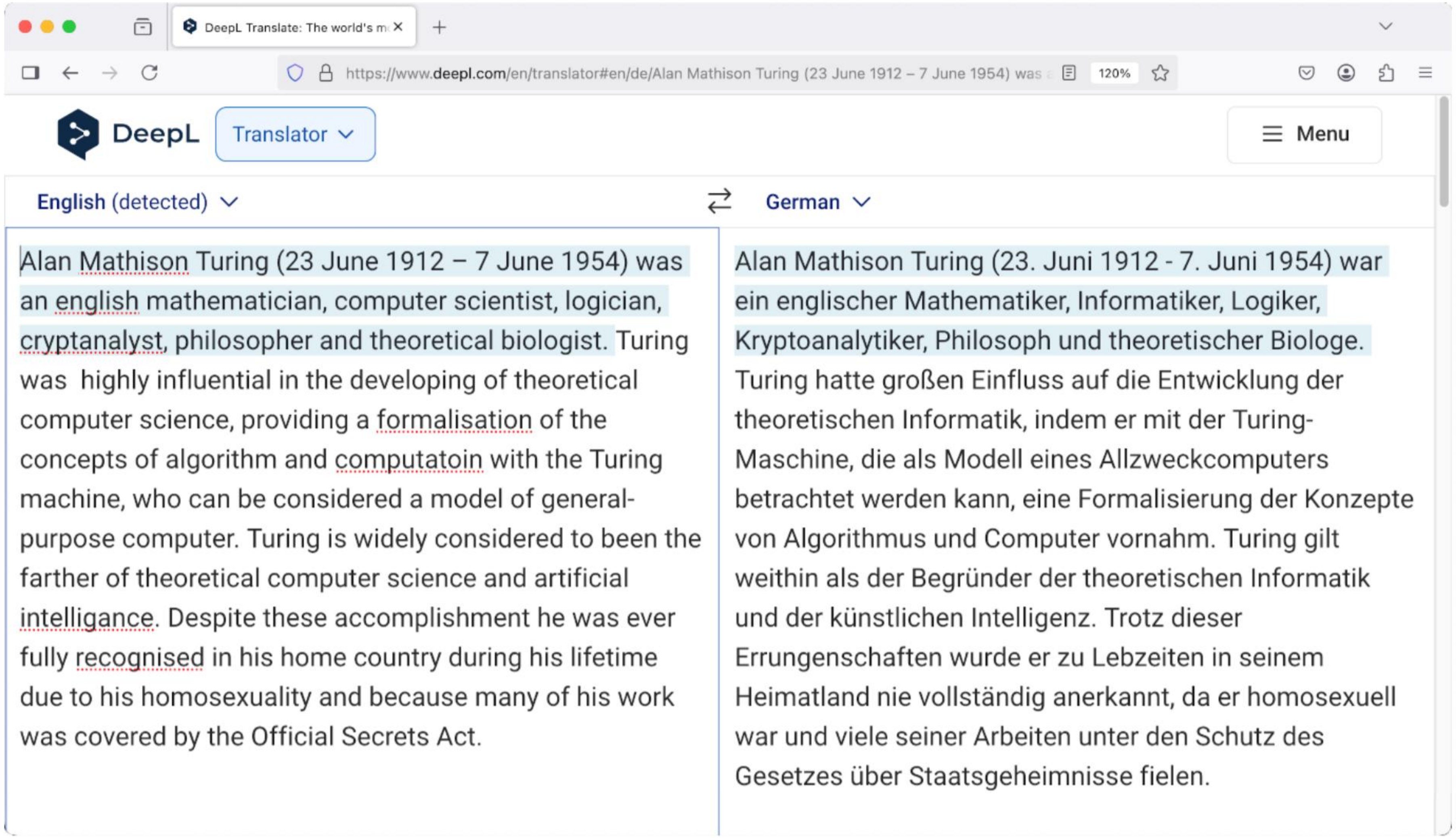




Optical Character Recognition

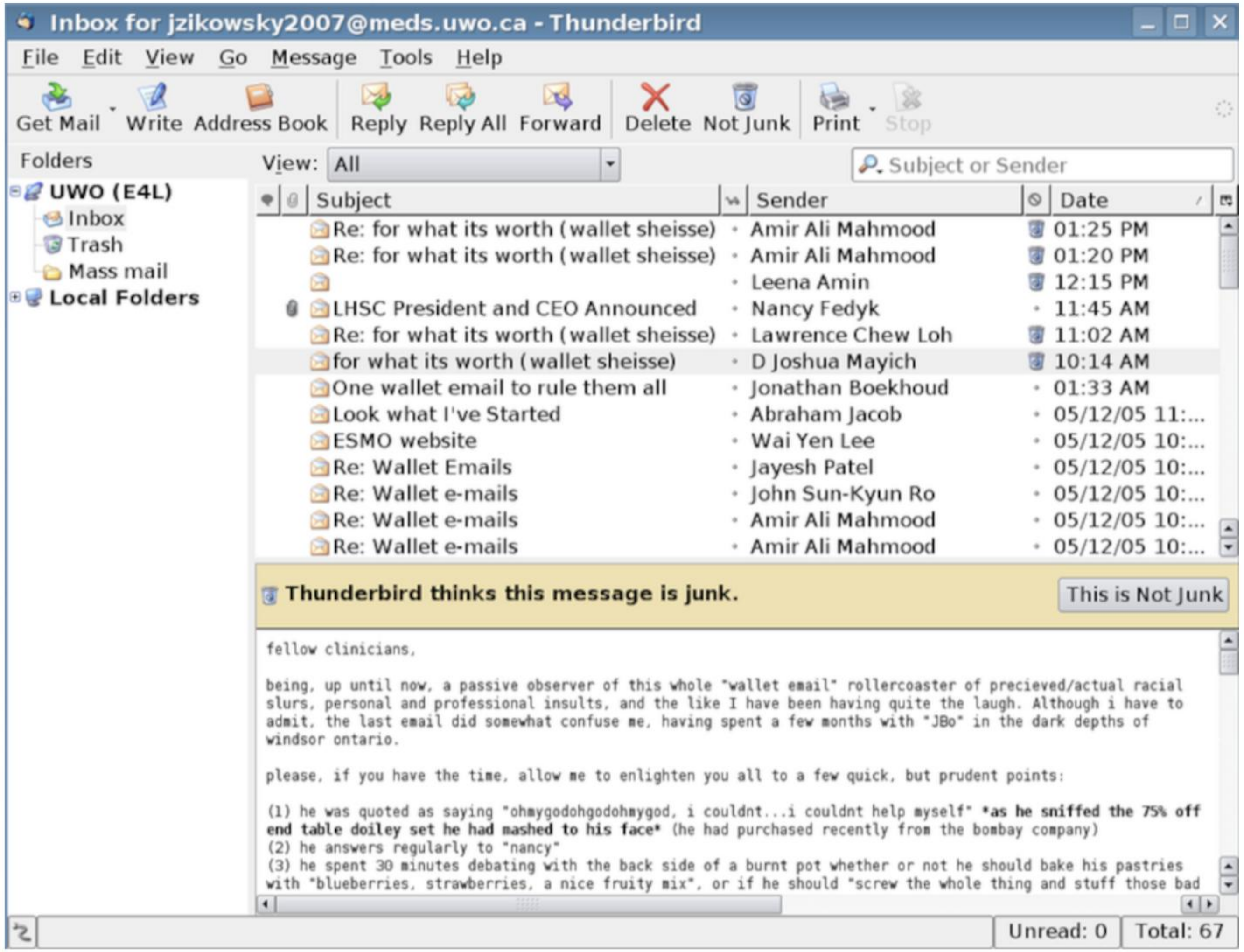


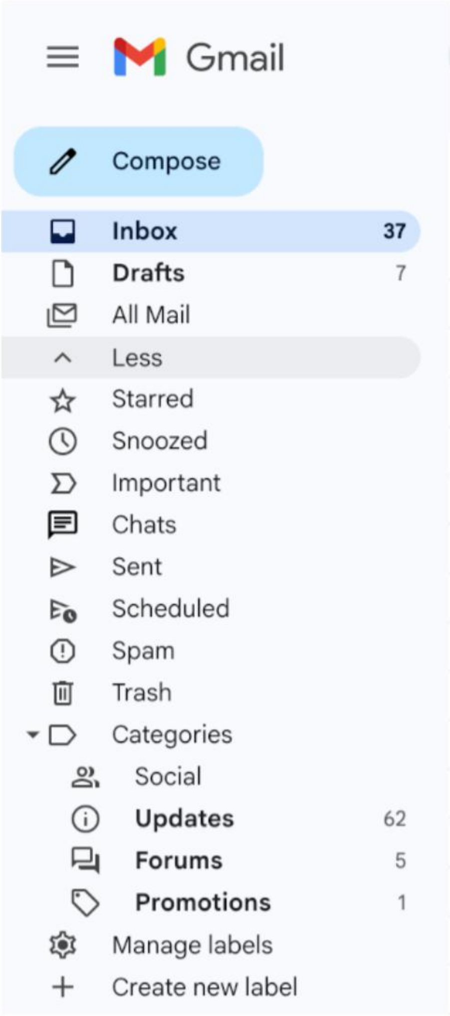
Machine Translation



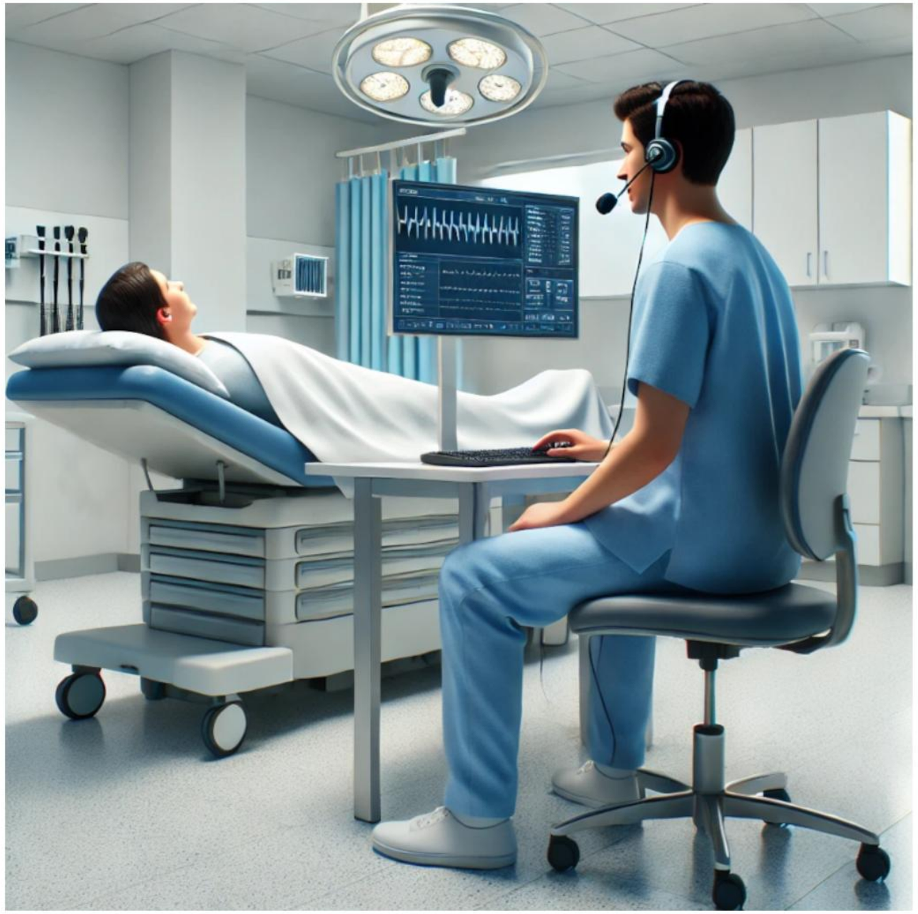
Machine Translation

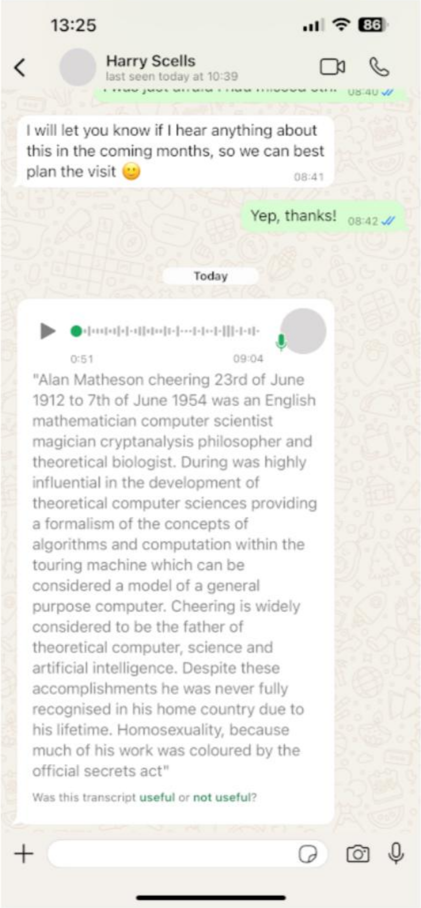
Spam Detection and Email Categorisation





Transcription and Virtual Assistants





Business Use Cases

- Customer service and customer relation management

  E.g., classification of customer support messages, dialogue systems and chatbots on web pages, or template-driven response generation.

- Business analytics and decision support

  E.g., sentiment analysis of company- or product-related news or social media, summarization of user feedback, market forecast analysis for investments.

- Legal compliance and moderation of user-generated content

  E.g., contract checking, invoice processing, classifying posts wrt. inappropriate content such as sexual messages or hate speech.

- Data journalism (e.g., on leaked documents)

  E.g., search, information extraction, summarization.

Governance Use Cases

- Public health monitoring

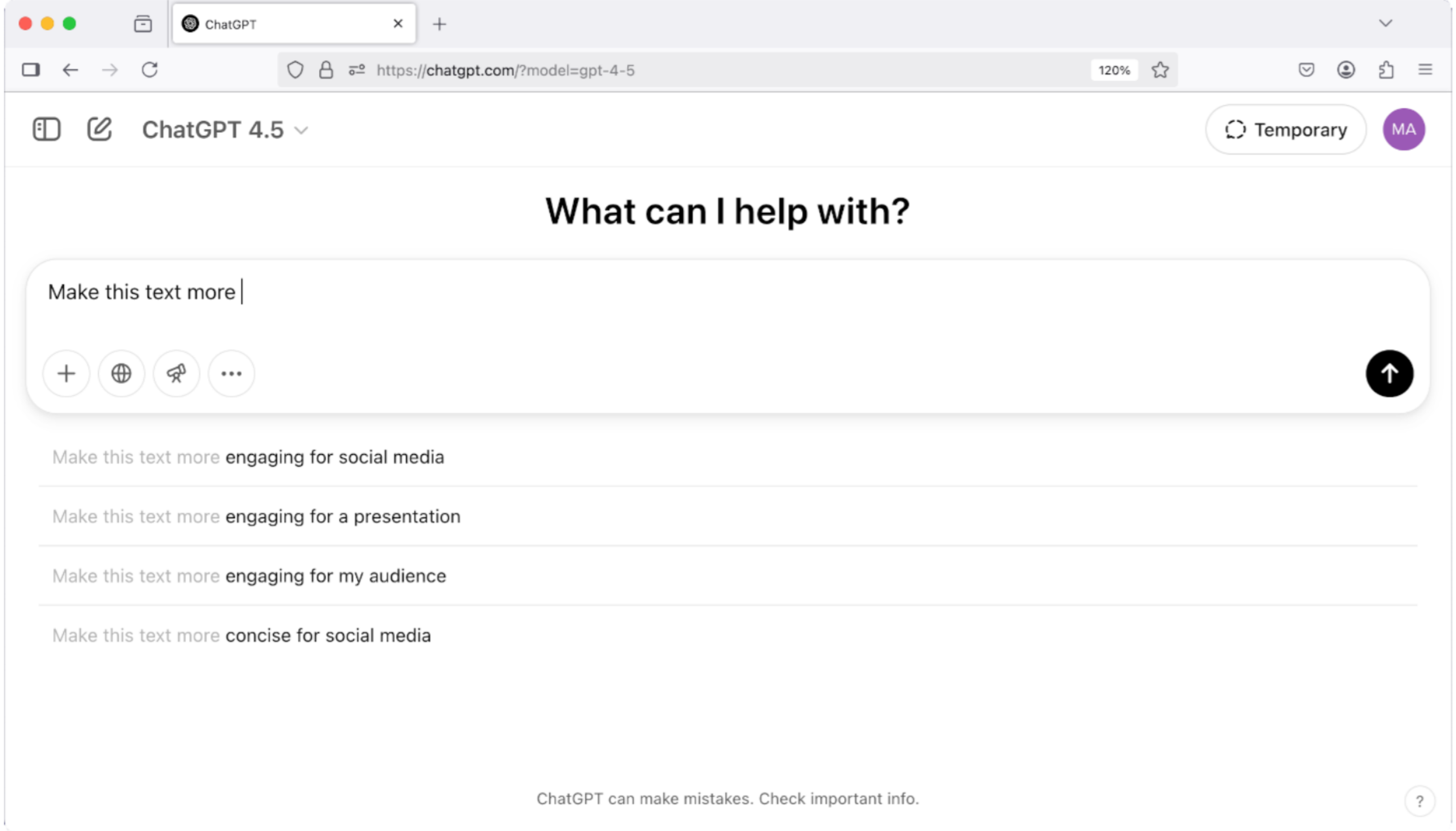
  E.g., social media monitoring for flu prediction
- Incident monitoring and disaster management

  E.g., social media monitoring for damage assessment
- Criminal investigation and counter terrorism

  E.g., style-based authorship analysis; search, classification, and summarisation of the dark web.
- Intelligence and espionage

  E.g., monitoring and transcription of communication worldwide.

Chatbots


ChatGPT
https://chatgpt.com/?model=gpt-4-5
120%
ChatGPT 4.5
Temporary
MA
What can I help with?
Make this text more
Make this text more engaging for social media
Make this text more engaging for a presentation
Make this text more engaging for my audience
Make this text more concise for social media
ChatGPT can make mistakes. Check important info.
?

Chatbots

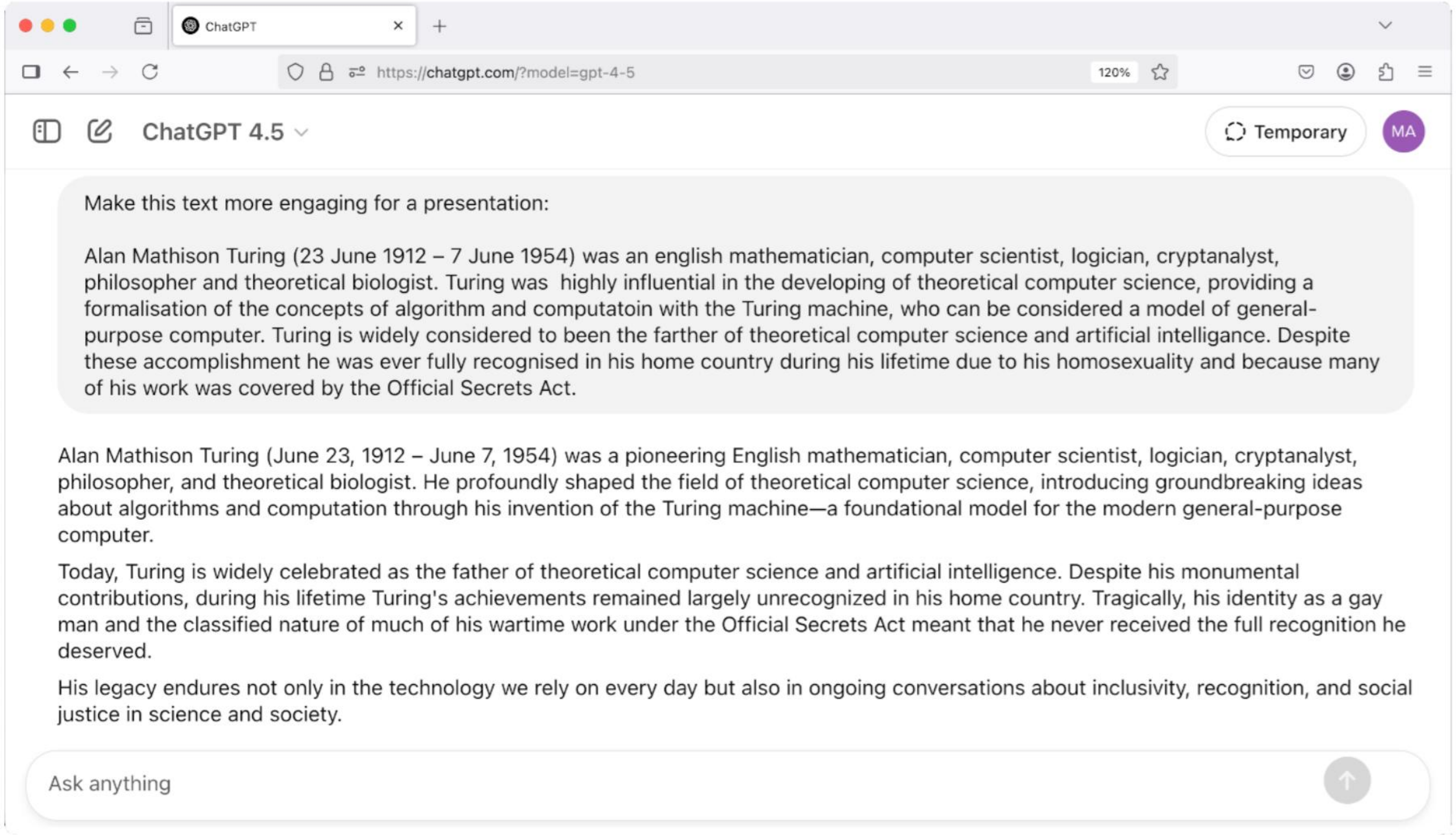
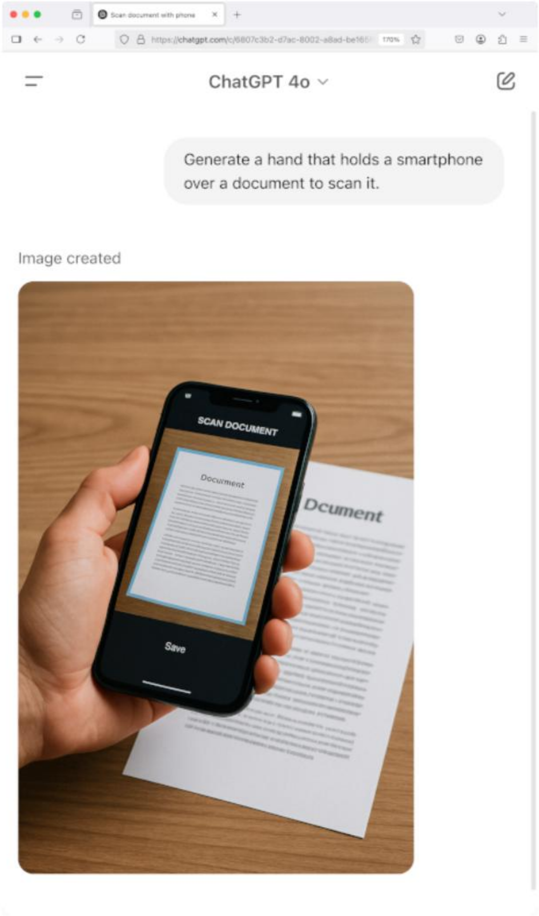
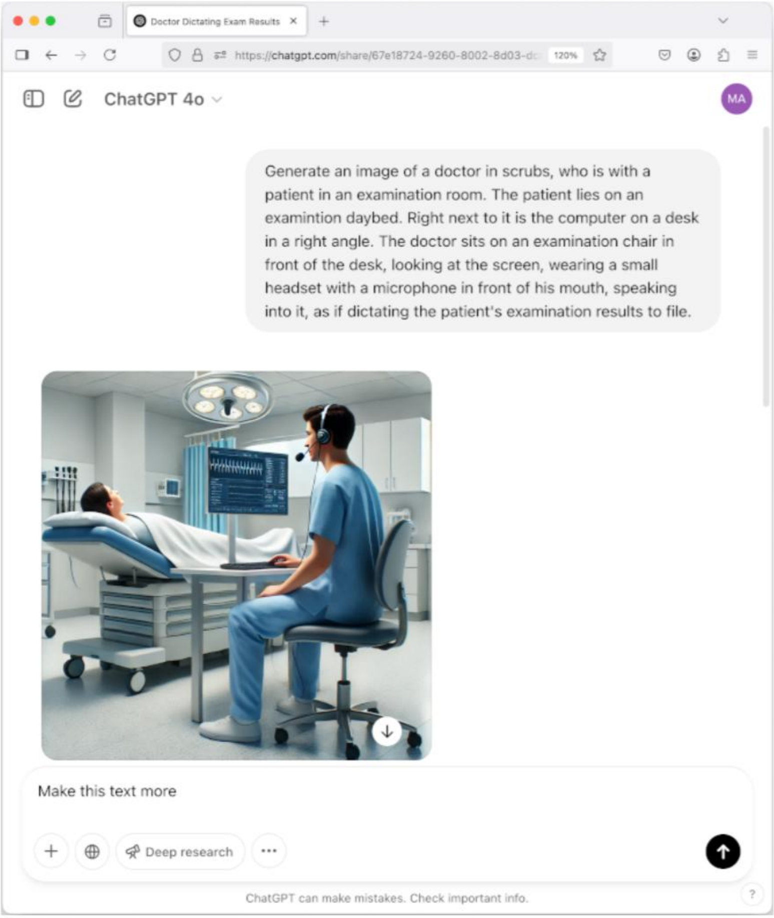

ChatGPT
https://chatgpt.com/?model=gpt-4-5
120%
ChatGPT 4.5
Temporary
MA
Make this text more engaging for a presentation:
Alan Mathison Turing (23 June 1912 – 7 June 1954) was an english mathematician, computer scientist, logician, cryptanalyst, philosopher and theoretical biologist. Turing was highly influential in the developing of theoretical computer science, providing a formalisation of the concepts of algorithm and computatoin with the Turing machine, who can be considered a model of general-purpose computer. Turing is widely considered to been the farther of theoretical computer science and artificial intelligance. Despite these accomplishment he was ever fully recognised in his home country during his lifetime due to his homosexuality and because many of his work was covered by the Official Secrets Act.
Alan Mathison Turing (June 23, 1912 – June 7, 1954) was a pioneering English mathematician, computer scientist, logician, cryptanalyst, philosopher, and theoretical biologist. He profoundly shaped the field of theoretical computer science, introducing groundbreaking ideas about algorithms and computation through his invention of the Turing machine—a foundational model for the modern general-purpose computer.
Today, Turing is widely celebrated as the father of theoretical computer science and artificial intelligence. Despite his monumental contributions, during his lifetime Turing's achievements remained largely unrecognized in his home country. Tragically, his identity as a gay man and the classified nature of much of his wartime work under the Official Secrets Act meant that he never received the full recognition he deserved.
His legacy endures not only in the technology we rely on every day but also in ongoing conversations about inclusivity, recognition, and social justice in science and society.
Ask anything

Image Generation from Text





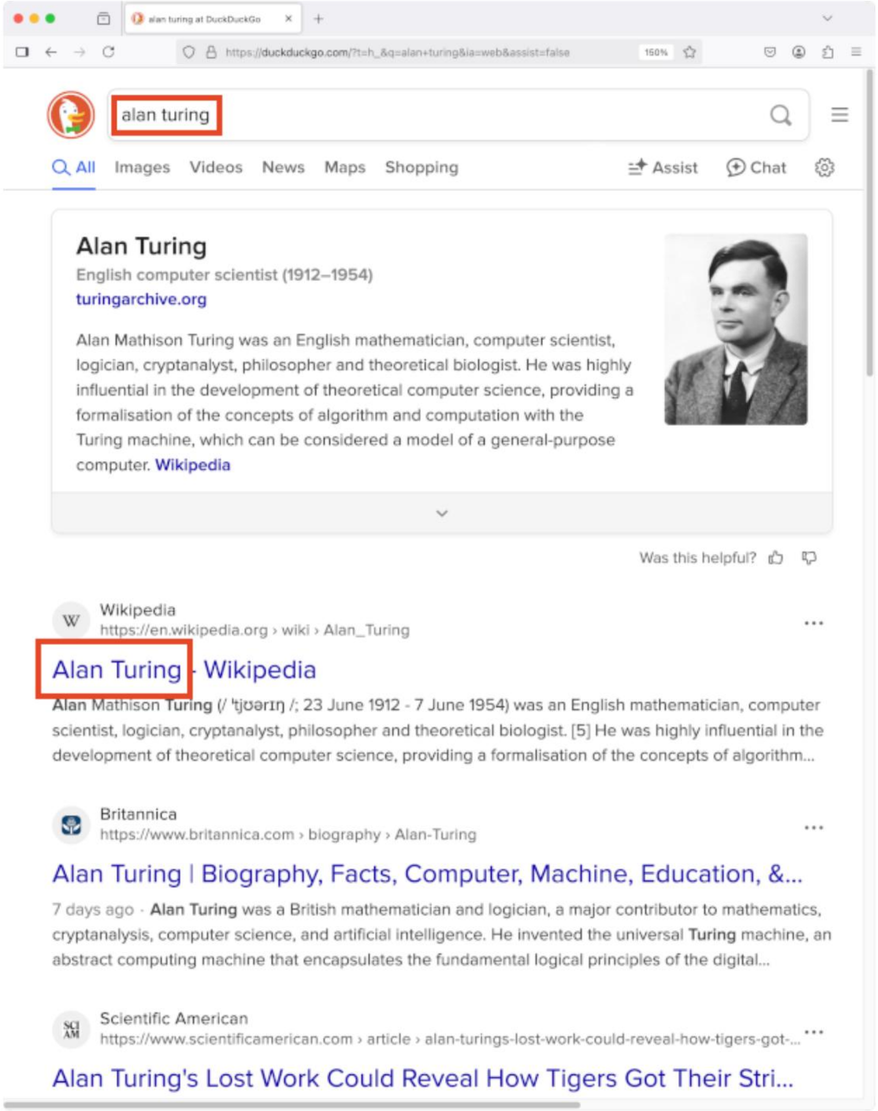
NLP Problems

Types of Tasks

- Analysis: the decoding of structured information from text
- Synthesis: the encoding of (structured) information into text

NLP Inside a Search Engine

- Search engines face the vocabulary mismatch problem between users and document authors.
- Users query one way, authors write another.
- Users still expect to find all relevant documents, regardless of how they write.
- Search engines implement a pipeline of basic NLP tasks to address this problem:
    - 1. Tokenization
    - 2. Lemmatization
    - 3. Stemming



Token and Sentence Splitting

```
Alan Mathison Turing (23 June 1912 - 7 June 1954) was¶
an English mathematician. He was highly influential¶
in the development of theoretical computer science,¶
providing a formalisation of the concepts of algorithm¶
and computation with the Turing machine, which can be¶
considered a model of a general-purpose computer.
```

Token and Sentence Splitting

Given a text as a character sequence

```
Alan Mathison Turing (23 June 1912 - 7 June 1954) was¶
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providing a formalisation of the concepts of algorithm¶
and computation with the Turing machine, which can be¶
considered a model of a general-purpose computer.
```

Token and Sentence Splitting

Given a text as a character sequence, split it into tokens.

Alan		Mathison		Turing		(	23		June		1912		-		7		June		1954	)		was	¶
an		English		mathematician	.		He		was		highly		influential	¶
in		the		development		of		theoretical		computer		science	,	¶
providing		a		formalisation		of		the		concepts		of		algorithm	¶
and		computation		with		the		Turing		machine	,		which		can		be	¶
considered		a		model		of		a		general-purpose		computer	.

Token and Sentence Splitting

Given a text as a character sequence, split it into sentences.

```
Alan Mathison Turing (23 June 1912 - 7 June 1954) was¶
an English mathematician. He was highly influential¶
in the development of theoretical computer science,¶
providing a formalisation of the concepts of algorithm¶
and computation with the Turing machine, which can be¶
considered a model of a general-purpose computer.
```

Remarks

- The goal of tokenization is typically to identify the words in a text.
- A word is defined as “an element or unit of speech [or] language”. “Any of the sequences of one or more sounds or morphemes (intuitively recognized by native speakers as) constituting the basic units of meaningful speech used in forming a sentence or utterance in a language (and in most writing systems normally separated by spaces); a lexical unit other than a phrase or affix; an item of vocabulary, a vocable.”[Oxford English Dictionary]
- In semiotics, a token is “a particular and individual sign, as opposed to the type of which it is an instance”.[Oxford English Dictionary] Non-words can also be tokens (e.g., punctuation, number, abbreviation, etc.).
- Tokens are distinguished from types; the so-called type–token distinction. Example: “A rose is a rose is a rose.” In one sense of ‘word’, this sentence contains three words, but in another sense of ‘word’, it contains eight words. In the first sense, the words are types, and in the second sense, they are tokens. In simplified terms, a token may refer to a particular instance of a word printed on a page, while a type refers to the abstract form it instantiates. [Wetzel 2006]
- Splitting a text into a sequence of tokens is typically the first step in solving a real-world problem like suggesting spelling corrections using natural language processing.

Stemming and Lemmatisation

Given a tokenised text, normalise every word token by replacing it with its lemma (CoreNLP lemmatiser)

be

Alan Mathison Turing (23 June 1912 - 7 June 1954) was¶

a he be

an English mathematician. He was highly influential¶

in the development of theoretical computer science,¶

provide concept

providing a formalisation of the concepts of algorithm¶

and computation with the Turing machine, which can be¶

consider

considered a model of a general-purpose computer.

Stemming and Lemmatisation

Given a tokenised text, normalise every word token by replacing it with its stem (Porter stemmer)

alan	mathison	ture			june				june				
Alan	Mathison	Turing	(	23	June	1912	-	7	June	1954	)	was	¶

	english			he		highli	influenti	
an	English	mathematician	.	He	was	highly	influential	¶

		develop		theoret	comput	scienc		
in	the	development	of	theoretical	computer	science	,	¶

provid		formalis			concept			
providing	a	formalisation	of	the	concepts	of	algorithm	¶

	comput			ture	machin					
and	computation	with	the	Turing	machine	,	which	can	be	¶

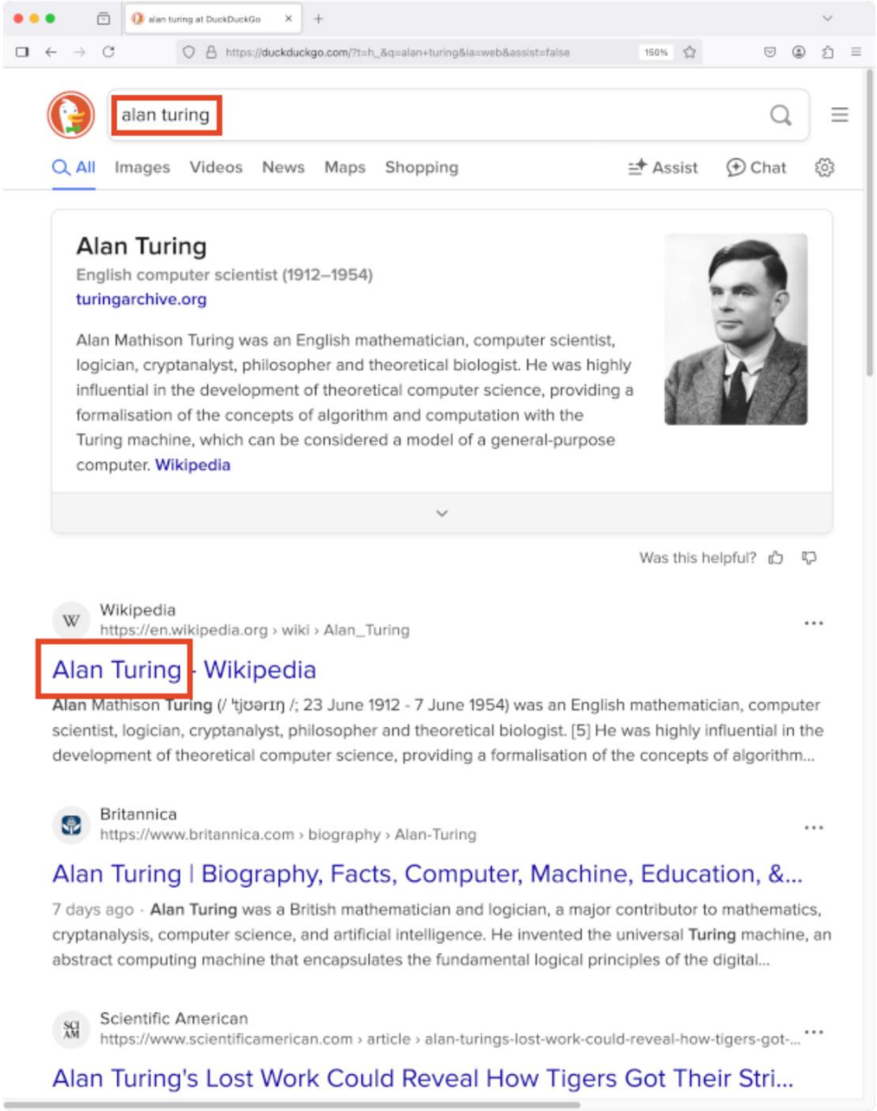
consid					gener		purpos	comput	
considered	a	model	of	a	general	-	purpose	computer	.

Remarks

- Lemmatisation is the task of reducing a word token to its lemma, “a lexical item as it is presented, usually in a standardized form, in a dictionary entry; a definiendum”. [Oxford English Dictionary]
- Stemming is the task of reducing a word token to its stem, “that part of an inflected word that remains unchanged in the process of inflection”. [Oxford English Dictionary]
- Lemmatisation exchanges irregular inflections, while stemming only removes affixes from a word token.
- Stemming is typically based on a set of rules serving as heuristics to remove inflections from word tokens. These rules may fail to map word tokens with a common stem to that stem, and they may also map word tokens with lexicographically different stems to the same stem candidate.
- The goals of lemmatization and stemming are to ease semantic matching in downstream steps when solving a real-world problem.

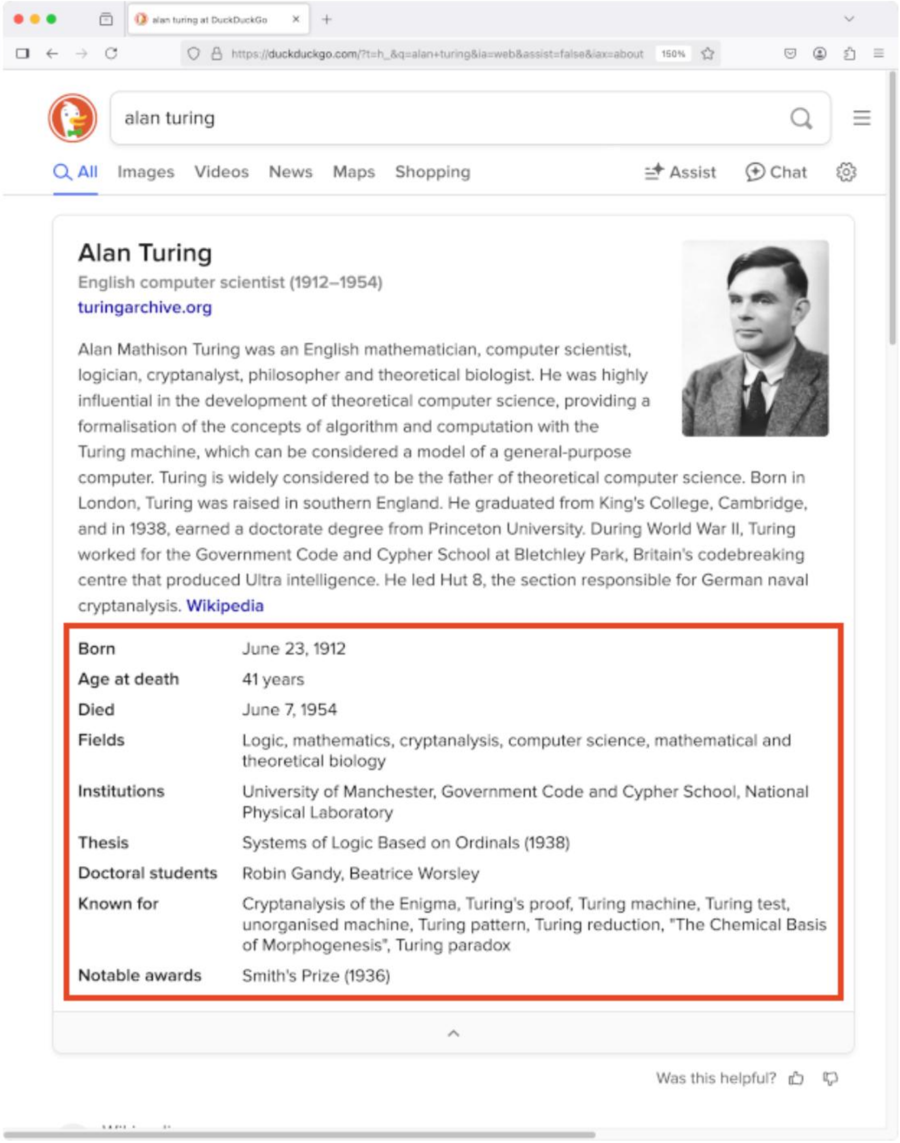
NLP Inside a Search Engine

- Search engines face the vocabulary mismatch problem between users and document authors.
- Users query one way, authors write another.
- Users still expect to find all relevant documents, regardless of how they write.
- Search engines implement a pipeline of basic NLP tasks to address this problem:
  - 1. Tokenization
  - 2. Lemmatization
  - 3. Stemming



NLP Inside a Search Engine

- Search engines return “knowledge panels” along search results.
- This relieves users from browsing the search results for basic facts.
- Unless it is collected manually, knowledge can be extracted from text using this pipeline:
  - 1. Tokenization
  - 2. POS tagging
  - 3. Named entity recognition
  - 4. Reference resolution
  - 5. Entity relation extraction



Part-of-Speech Tagging

Given a tokenised text, identify the part of speech of each token using Universal POS tags

PROPN	PROPN	PROPN	SYM	NUM	PROPN	NUM	SYM	NUM	PROPN	NUM	SYM	VERB
Alan	Mathison	Turing	(	23	June	1912	-	7	June	1954	)	was¶

DET	ADJ	NOUN	PUNCT	PRON	VERB	ADV	ADJ
an	English	mathematician	.	He	was	highly	influential¶

ADP	DET	NOUN	ADP	ADJ	NOUN	NOUN	PUNCT
in	the	development	of	theoretical	computer	science	.

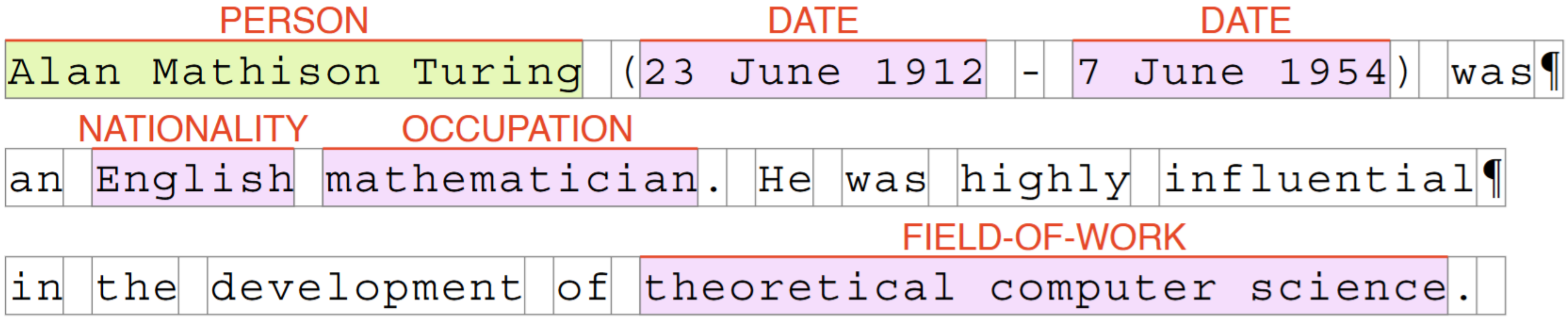
ADJ: adjective
ADP: adposition
ADV: adverb
AUX: auxiliary
CCONJ: coordinating conjunction

DET: determiner
INTJ: interjection
NOUN: noun
NUM: numeral
PART: particle
PRON: pronoun

PROPN: proper noun
PUNCT: punctuation
SCONJ: subordinating conjunction
SYM: symbol
VERB: verb
X: other

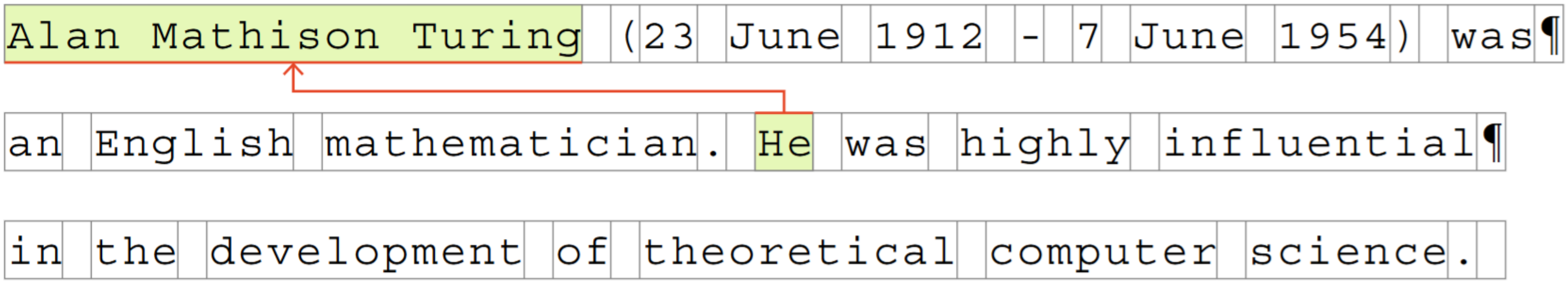
Named Entity Recognition

Given a POS-tagged text, identify token sequences that name real-world objects.



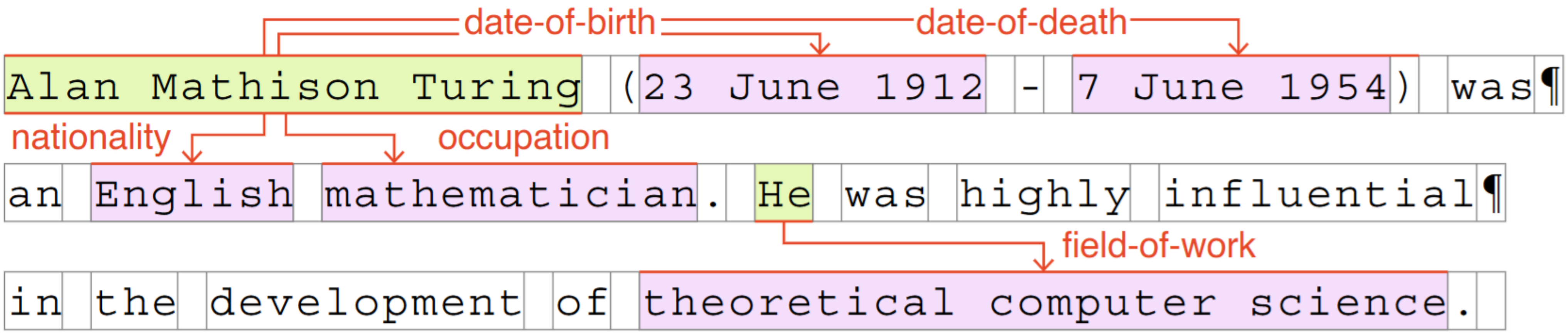
Coreference Resolution

Given a POS-tagged text, map entities to their coreferents.



Entity Relation Extraction

Given a NER-tagged text with coreferences, identify relations among entities.



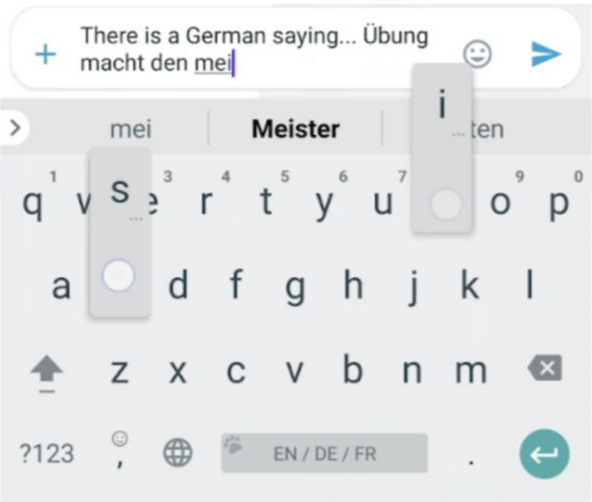
The goal of entity relation extraction is to identify knowledge triples of the form (subject, predicate, object) which form the basis of modern knowledge bases.

Types of Tasks

- Analysis: the decoding of structured information from text
- Synthesis: the encoding of (structured) information into text

Autocomplete

- Typing on mobile devices is relatively slow.
- Many users are further slowed down by the use of logographic language and/or disabilities.
- A key input method includes predicting the currently typed word and/or the next word.
- Today, many systems construct a statistical language model for this purpose as follows:
  - 1. Corpus construction
  - 2. Tokenization
  - 3. Token k-gram counting
  - 4. Next token prediction



Token n-gram Counting

Given a tokenised text, tabulate its token n-gram counts, where a n-gram is a token sequence of length n.

Alan Mathison Turing (23 June 1912 - 7 June 1954) was¶
an English mathematician. He was highly influential¶
in the development of theoretical computer science.

Token n-gram Counting

Given a tokenised text, tabulate its token unigram counts, where a unigram is a token sequence of length 1.

Alan Mathison Turing (23 June 1912 - 7 June 1954) was¶
an English mathematician. He was highly influential¶
in the development of theoretical computer science.

Token n-gram Counting

Given a tokenised text, tabulate its token bigram counts, where a bigram is a token sequence of length 2.

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Alan Mathison Turing (23 June 1912 - 7 June 1954) was¶
an English mathematician. He was highly influential¶
in the development of theoretical computer science.
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in the development of theoretical computer science.
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Token n-gram Counting

Given a tokenised text, tabulate its token trigram counts, where a trigram is a token sequence of length 3.

Alan Mathison Turing (23 June 1912 - 7 June 1954) was¶
an English mathematician. He was highly influential¶
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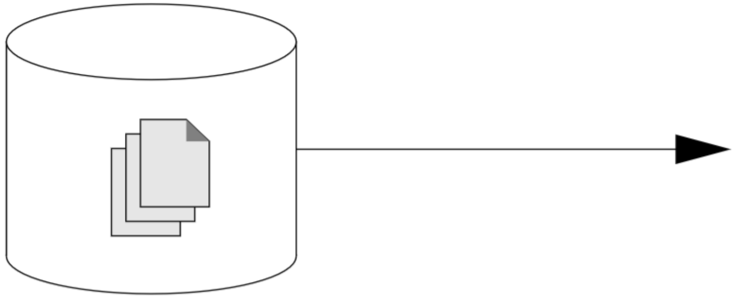
Token n-gram counting

Given a tokenised corpus, tabulate its token n-gram counts.



3-gram	Count
`Alan Mathison Turing`	2,800
`Mathison Turing (`	52
`Turing ( 23`	<42
⋮	⋮

How scalable is token n-gram counting?

Token n-gram counting

1-gram	Count
Alan	24,000,000
Mathison	110,000
Turing	1,300,000
(	3,300,0000,000
23	320,000,000
June	210,000,000
1912	12,000,000
–	19,000,000,000
7	760,000,000
1954	20,000,000
)	3,300,0000,000
was	3,700,000,000
an	2,500,000,000
English	270,000,000
mathematician	2,200,000
.	39,000,000,000
He	2,400,000,000
highly	73,000,000
influential	9,300,000
in	15,000,000,000
the	43,000,000,000
development	420,000,000
of	25,000,000,000
theoretical	25,000,000
computer	240,000,000
science	260,000,000

2-gram	Count
Alan Mathison	4,500
Mathison Turing	3,000
Turing (	1,400
(23	330,000
23 June	670,000
June 1912	51,000
1912 –	190,000
– 7	870,000
7 June	550,000
June 1954	100,000
1954)	640,000
) was	41,000,000
was an	41,000,000
an English	6,000,000
English mathematician	36,000
mathematician .	130,000
. He	320,000,000
He was	270,000,000
was highly	1,600,000
highly influential	170,000
influential in	750,000
in the	3,500,000,000
the development	57,000,000
development of	78,000,000
of theoretical	1,500,000
theoretical computer	300,000
computer science	12,000,000
science .	11,000,000

3-gram	Count
Alan Mathison Turing	2,800
Mathison Turing (	52
Turing (23	<42
(23 June	883
23 June 1912	612
June 1912 –	1,400
1912 – 7	92
– 7 June	737
7 June 1954	1,100
June 1954)	1,900
1954) was	109
) was an	9,100
was an English	170,000
an English mathematician	7,600
English mathematician .	1,100
mathematician . He	7,400
. He was	47,000,000
He was highly	110,000
was highly influential	12,000
highly influential in	28,000
influential in the	220,000
in the development	8,500,000
the development of	44,000,000
development of theoretical	19,000
of theoretical computer	30,000
theoretical computer science	310,000
computer science .	540,000

Questions?